MILITARY SPECIFICATION

TEST SETS, MICROWAVE CALIBRATION: AN/USM-129, AN/USM-129A, AN/USM-129B

1. SCOPE

1.1 Scope. This specification covers mobile test sets known as Test Set, Microwave Calibration: AN/USM-129, AN/USM-129A or AN/USM-129B, and hereinafter designated in this specification as "MCTS." Each set consists of a series of Universal Microwave Calibrators and associated accessory equipments, stored in combination cases, which is capable of calibrating microwave test equipment for RF power output, voltage standing wave ratio (VSWR), attenuation and frequency response in the frequency range of 0.3 milomegacycles per second (kmcps) to 40.0 kmcps.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-45208	Inspection Requirements: General Specification for
MIL-U-46351 (ORD)	Universal Microwave Calibrators
MIL-F-46352 (ORD)	Frequency Multipliers

DRAWINGS

ORDNANCE CORPS

A7920100	Heterodyne Frequency Meter 100—10000MC
A7920260	Combination of Adopted Items, AN/USM-129B.
A10519341	Calorimeter
A10519351	Calorimeter
A10519401	Amplifier, Standing Wave
A10519450	Signal Generator, Stabilized
A10519472	Attenuator Calibrator Assembly
A10528291	Receptacle, Female
A10528410	Combinations of Adopted Items, AN/USM-129A.
A10528591	Power Meter Amplifier, Calorimetric
A10528620	Combinations of Adopted Items, AN/USM-129

PUBLICATIONS

ORDNANCE CORPS

TM 9-4931-15/3	Operator, Organizational, Field and Depot Maintenance Manual (Calibration Procedures)
TM- . . . . . 0 . . .	

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial Production Sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the

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same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction. The MCTS shall be manufactured in accordance with the drawings in 2.1, as required, and all drawings referenced thereon. The universal microwave calibrators and frequency multipliers which form a part of the MCTS shall comply with the appropriate specifications in 2.1. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired electrical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.4 Electrical. The MCTS with 117 volts AC ± 10 percent, 60 ± 5 cps current applied shall be capable of calibrating test equipment for the appropriate parameters within the ranges and accuracies indicated in the following paragraphs.

3.4.1 *Frequency Range.* The frequency range of the MCTS shall be from 0.3 kmcps to 40.0 kmcps.

3.4.2 *Output Power.*

3.4.2.1 *Average RF Power.* The MCTS shall be capable of measuring average RF power from 0.1 milliwatt to 1.0 watt to an accuracy of ± 4.0 percent in the frequency range 0.3 kmcps to 10.0 kmcps and ± 5.0 percent from 10.0 kmcps to 40.0 kmcps except that at checkpoints (9.0 kmcps and 11.2 kmcps) the accuracy shall be ± 3.0 percent. These accuracies are exclusive of instrumentation error.

3.4.2.2 *Peak Pulse Power.* The MCTS shall be capable of measuring peak pulse power to an accuracy of ± 1.0 decibel (db) with a minimum pulse width of 0.5 microseconds and a pulse recurrence frequency of 200.

3.4.3 *VSWR.* The MCTS shall be capable of measuring VSWR up to a value of 4.0 with an accuracy of ± 0.05 and from 4.0 to 10.0 with an accuracy of ± 0.1.

3.4.4 *Attenuation.* The MCTS shall be capable of measuring the attenuation of an external attenuator up to 40.0 db with an accuracy of ± 0.5 db and an internal attenuator up to 105 db with an accuracy of ± 2.0 percent.

3.4.5 *Frequency Measurement.*

3.4.5.1 *Frequency Measurement of an RF Signal.* The MCTS shall be capable of measuring the frequency of an RF signal to an accuracy of ± 0.25 percent.

3.4.5.2 *Frequency Response of a Passive Device.* The MCTS shall be capable of measing the frequency response of a passive device to an accuracy of ± 0.03 percent.

3.5 Environmental.

3.5.1 *Operating Tests.* At the start and conclusion of each operating test, the equipment shall be tested to determine whether the requirements of 3.4 are met.

3.5.1.1 *Temperature.* The equipment shall be tested while exposed and thermally stabilized at ambient temperatures of plus 32 degrees and plus 110 degrees Fahrenheit.

3.5.1.2 *Humidity.* The equipment shall be exposed to an atmosphere of 95 percent relative humidity at plus 77 degrees Fahrenheit for 24 hours in non-operating condition. At the end of this period, the equipment shall be operated for eight hours at the same temperature and relative humidity.

3.5.1.3 *Altitude.* The equipment shall be capable of operating at a reduced pressure of 20.58 inches of mercury (equal to an altitude of 10,000 feet) for four hours.

3.5.2 *Non-operating Tests.* At the start and conclusion of each non-operating test, the equipment shall be tested at normal conditions for compliance with the requirements of 3.4.

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3.5.2.1 *Temperature.* The equipment shall be stored for four hours while exposed and thermally stabilized at ambient temperatures of minus 25° and plus 158° Fahrenheit.

3.5.2.2 *Altitude.* The equipment shall be stored for one hour at a reduced pressure of 3.44 inches of mercury (equal to an altitude of 50,000 feet).

3.5.2.3 *Normal Conditions.* For normal conditions, the temperature shall be 77° Fahrenheit and the relative humidity shall be 60 percent at sea level.

3.6 Workmanship. The MCTS, including all parts and accessories, shall be manufactured and finished in a thoroughly workmanlike manner and in accordance with good commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and test shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General Requirements. Inspection shall be in accordance with Specification MIL–I–45208.

4.3 Initial Production Sample. Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 *Production Sample Failure.* Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Material. Control of material shall be in accordance with test data requirements of Specification MIL–I–45208.

4.5 Inspection of Components. The contractor shall ascertain that prior to assembly all components procured under separate specifications or drawings, have been inspected, tested and accepted in accordance with their respective specifications or drawings.

4.6 Electrical.

4.6.1 *Frequency Range.* Each of the universal microwave calibrators forming a part of the MCTS shall comply with the frequency range requirements of MIL–U–46351(ORD).

4.6.2 *Average RF Power.* When the measurement of average RF power is performed in accordance with the procedure and equipment given in the Publications listed in 2.1 and the calorimeters (drawings A10519341 and A10519351) and calorimetric power meter amplifier (drawing A10528591) are replaced in the circuit by a precision standard power meter, the measurement shall repeat within the accuracy given in 3.4.2.1.

4.6.3 *Peak Pulse Power.* When the measurement of peak pulse power is performed in accordance with the procedure and equipment given in the publications listed in 2.1 and the calorimeters (drawings A10519341 and A10519351) and calorimetric power meter amplifier (drawing A10528591) are replaced in the circuit by a precision standard power

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meter, the measurement shall repeat within the accuracy given in 3.4.2.2.

4.6.4 *VSWR.* When the measurement of VSWR is performed in accordance with the procedure and equipment given in the publications listed in 2.1, and the standing wave amplifier (drawing A10519401) is replaced in the circuit by a precision standard standing wave amplifier, the measurement shall repeat within the accuracy given in 3.4.3.

4.6.5 *Attenuation.* When the measurement of attenuation is performed in accordance with the procedure and equipment given in the publications listed in 2.1 and the attenuation calibrator (drawing C10519472) is replaced in the circuit by a precision standard attenuator, the measurement shall repeat within the accuracy given in 3.4.4.

4.6.6 *Frequency of an RF Signal.* When the measurement of the frequency of an RF signal is performed in accordance with the procedure and equipment given in the publications listed in 2.1 and the heterodyne frequency meter (drawing A7920100) is replaced in the circuit by a precision standard frequency meter, the measurement shall repeat within the accuracy given in 3.4.5.1.

4.6.7 *Frequency Response of a Passive Device.* When the measurement of the frequency response of a passive device is performed in accordance with the procedure and equipment given in the publications listed in 2.1 and the stabilized signal generator (drawing A10519450) and the heterodyne frequency meter (drawing A7920100) are replaced by a precision standard signal generator and frequency meter respectively, the measurement shall repeat within the accuracy given in 3.4.5.2.

4.7 Environmental.

4.7.1 *Operating Tests.*

4.7.1.1 *Temperature.* When tested while exposed and thermally stabilized at ambient temperatures of plus 32° and plus 110° Fahrenheit, the MCTS shall conform to the requirements of 3.4.

4.7.1.2 *Humidity.* After having been exposed to an atmosphere of 95 percent relative humidity at plus 77° Fahrenheit for 24 hours in non-operating condition, the MCTS, when tested while exposed to the same temperature and relative humidity, shall comply with the requirements of 3.4 for eight hours of continuous operation.

4.7.1.3 *Altitude.* When tested while exposed to a reduced pressure of 20.58 inches of mercury (equal to an altitude of 10,000 feet), the MCTS shall comply with the requirements of 3.4 for four hours of continuous operation.

4.7.2 *Non-operating Tests.*

4.7.2.1 *Temperature.* After having been stored for four hours while exposed and thermally stabilized at an ambient temperature of minus 25 degrees Fahrenheit, the MCTS shall be returned to normal conditions and tested for compliance with 3.4. The test shall be repeated at an ambient temperature of plus 158 degrees Fahrenheit.

4.7.2.2 *Altitude.* After having been stored for one hour while exposed to a reduced pressure of 3.44 inches of mercury (equal to an altitude of 50,000 feet), the MCTS shall be returned to normal conditions and tested for compliance with 3.4.

4.8 Sampling.

4.8.1 *Environmental Tests.* The environmental tests of 4.7 shall be performed on a number of representative subassemblies of the MCTS as specified by the contracting officer.

4.8.2 *Electrical Tests.* The electrical tests of 4.6 shall be performed on each MCTS.

5. PREPARATION FOR DELIVERY

5.1 Packaging. No further packaging shall be required in addition to the combination cases in which the MCTS is housed. At least one of the two latches on each side of each combination case shall be secured with a security seal prior to shipment.

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6. NOTES

6.1 Intended Use. The MCTS covered by this specification is used to perform the following measurements in the frequency range of 0.3 to 40.0 kmcps:

(a) The frequency measurement of an active source.

(b) The relative frequency response of a passive device.

(c) The peak power measurement of a pulsed signal source.

(d) The measurement of average power, attenuation and voltage standing wave ratio.

6.2 Ordering Data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Applicable stock number.

(c) Provisions for initial production sample, including date of delivery (see 3.1).

(d) Environmental tests required (see 4.8.1).

6.3 Design Information. Design information regarding specific equipment not covered or referenced by this Specification may be obtained from the procuring activity upon specific request of the contractor.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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